

ITCC103 (IT FUNDAMENTALS) Week II Assignment

Module 1 - Unit 1 (Common Computing Devices)

Subject Code-Section: ITCC103-102I

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Questions:

1. Based on the different operating systems mentioned (Windows, macOS, Linux, Android, iOS), how do you think a user's choice of OS for a personal computer or mobile device impacts their daily digital life and the applications they can use?

Answer: Choosing an OS depends on what they need, it should be aligned to what the user wants it to use for. Example is if they just want it for work and general task, they should use windows OS, if they choose like linux, it might be harder for them to use it because it typically use and useful for programmers and techy people.

2. Explain the difference between a Graphical User Interface (GUI) and a Command-Line Interface (CLI). In what specific scenarios might a user prefer to use a CLI over a GUI, even though a GUI is more common for personal use?

Answer: GUI is much better than CLI in any scenario because its easier for typical users to navigate through an app or software. GUI also good for media displaying like photos and videos. While as CLI is text-based and uses commands to run things, it is much faster or lightweight to load than a GUI because it does not need to render media.

3. Our lesson describes the boot process and how the OS loads after the Power-On Self-Test (POST). Describe this process in your own words, and explain what you think would happen if a critical part of the POST failed.

Answer: POST has a process of checking every components installed in the unit. It has a list of it and it analyze or check each installed things if they do work. If somehow the POST process detects something, it might or will display an error message in the monitor and you will not be able to use your PC. What I thought is if you don't have experience in computer parts, you might need to call for help so that they can diagnose the problem and solve it.

4. What is multitasking, and how do operating systems like Windows or macOS enable it? Describe a situation where you have used multitasking to improve your efficiency, and explain which OS features helped you do so.

Answer: Multitasking is a process of doing all works at the same time. Operating Systems has ways to make our life easier by utilizing tab controls and commands in the keyboard like copy pasting. A scenario example is when making a website, you use switch tabs between an Code editor, the search engine, file managers and more.

5. Explain the concept of device drivers. Why are they essential for an operating system, and what might happen if a device driver for a new piece of hardware, like a printer or a graphics card, is not installed or is outdated?

Answer: Device drivers are software that allow the operating system to communicate with hardware devices like printers, graphics cards, and keyboards. They are essential because they help the OS recognize and control the hardware. If a driver is missing or outdated, the device may not work properly, have limited features, throws errors, or cause compatibility and stability issues.

6. The file system is responsible for organizing data on a storage device. Describe the hierarchical structure of a file system and explain why this structure is beneficial for a user trying to manage their files and folders.

Answer: File systems are organized like an upside-down tree: root > folders > subfolders > files. This helps users group related files, avoid naming conflicts, and easily find and manage files.

7. What is the purpose of a user account in an operating system? Discuss the difference between a standard user account and an administrator account, and explain why it is a good security practice to primarily use a standard account for daily tasks.

Answer: A user account identifies who's using the device and sets their permissions. Standard accounts can use programs and save files but can't change system settings. Admin accounts have full control. Using a standard account daily is safer, if malware gets in, it's limited to what that account can do.

8. The material mentions file and folder permissions. Explain what file permissions are and provide a real-world example of a situation where setting specific permissions (e.g., read-only access) would be important.

Answer: Permissions control who can view, edit, or delete a file. Example: a teacher shares a syllabus as read-only so students can view it but can't accidentally change or delete it.

9. Explain the role of the OS in managing system resources like the CPU, RAM, and storage. How does the OS prevent a single application from using all of a computer's resources and causing it to crash or freeze?

Answer: The OS allocates CPU time, RAM, and storage access among running programs. It prevents one app from hogging everything by giving each program small time slices and limiting how much memory it can use.

10. Our lesson discusses various types of utility programs like disk cleanup and defragmentation tools. Based on your understanding, describe a situation where you would use one of these tools and explain how it would improve your computer's performance.

Answer: If your computer is slow and storage is nearly full, Disk Cleanup removes temp files and clutter to free up space, helping the system run faster. Defragmentation can further help by rearranging scattered file data so the drive reads files more quickly.